

Round 1

Elementary Regional – 2022 Governor's Cup

1. A large angle measures 140 degrees and is divided into three smaller angles. Two of the smaller angles have the same measure and the third angle measures 50 degrees. What is the measure, in degrees, of each of the other two angles?

45 (DEGREES)
2. After leaving Harvard, this man licensed the MS-DOS operating system to IBM. With Paul Allen, he founded a company that eventually created the Xbox, and Windows. Who is this philanthropist and co-founder of Microsoft?

(BILL) GATES
3. Among this author's works were *Mother Goose in Prose* and *American Fairy Tales*. He set 14 books in a magical land that housed the Tin Woodman, Scarecrow, and Cowardly Lion. Identify the author of *The Wonderful Wizard of Oz*.

(L(YMAN) FRANK) BAUM
4. An early form of this was known as the gift economy. A form of this known as barter does not use money as part of the exchange. Give this term for an exchange of goods, services, and currency.

TRADE
5. This substance found in soil is brown or black, and usually contains 60 percent carbon. As this substance decomposes, plants can absorb its nutrients. Name this substance that is made up of decomposed plants and animals.

HUMUS
6. An author uses this part of speech to demonstrate a feeling that grabs the reader's attention. What is the part of speech of "ouch," "hey," and "wow"?

INTERJECTION(S)
7. Consider the pattern of numbers that starts 12, 60, 300, and so on, in which each number in the pattern is 5 times larger than the previous number. What is the next number in the pattern?

1500
8. In one country, war and union versions of this wind instrument existed. Basic versions have a chanter, drones, another valved tube, and a sac made of rubber or animal skin. What is this loud instrument associated with the Scottish Highlands, sometimes played at funerals?

BAGPIPE(S)
9. It takes 270 to reach a majority in this group. Each state is represented in this body equal to the number of its congressional delegation. What is this two-word body that elects the U.S. president?

ELECTORAL COLLEGE
10. They can be made from mRNA, inactivated viruses, or viral particles. Examples include the DPT and a recent one created by Moderna. What is this medical technology, used to prevent or weaken illnesses from viruses, and typically given through shots?

VACCINATION(S) (Accept VACCINES)

11. In Mrs. Reed's classroom, the ratio of her non-fiction books to fiction books is 2 to 5. If there are a total of 28 books in Mrs. Reed's classroom, how many of the books are fiction?
- 20 (BOOKS)
12. Spelling required. The teacher seldom offered his opinion on the students' group projects. Spell the word in the preceding sentence that is an antonym of "often."
- S-E-L-D-O-M
13. This word is both the name of an island off the coast of San Francisco, California and the prison that is located on it. Give the name of this notorious prison, sometimes called "The Rock".
- ALCATRAZ
14. Traditionally, this dance was performed to gourd drum, rhythm stick, and lava stone accompaniment. Dancers perform barefoot and wore necklaces, bracelets, anklets, and headpieces made of leaves or flowers. Identify this storytelling dance native to Polynesian islands, especially Hawaii.
- HULA
15. Two-word answer. This type of asexual reproduction is used by many bacteria. Identify this process, by which an organism makes an exact copy of itself and splits in half.
- BINARY FISSION
16. At age 15, this man was hired by the Mobile & Ohio Railroad. On April 30, 1900, he saved the life of his fireman but sacrificed his own. Name this railroad engineer who allegedly died with one hand on the train's whistle and the other on its brake.
- (JOHN LUTHER "CASEY") JONES
17. He became world famous for a flight that he made on his plane, the Spirit of St. Louis. He made the first nonstop flight from New York City to Paris in 1927. Name this famous pilot.
- (CHARLES) LINDBERGH
18. It is the planned target of NASA's Dragonfly mission. It has a thick atmosphere, and it is the second largest moon by volume in the solar system. Identify this largest moon of Saturn.
- TITAN
19. Properties of this element of art include saturation, brightness, and hue. They can be complementary, as well as primary, secondary, warm, and cool. What is this element, whose examples include purple and green?
- COLOR
20. State your answer as a fraction. A spinner is divided into ten equal sectors numbered one through ten. What is the probability of landing on a prime number?
- $\frac{2}{5}$
21. After being denied admission to multiple medical schools, this trailblazer was finally admitted to Geneva Medical School, graduating first in her class in 1849. Name this doctor, the first woman in the United States to graduate from medical school.

(ELIZABETH) BLACKWELL

22. Americans who helped negotiate this treaty were John Jay, Benjamin Franklin, Henry Laurens, and John Adams. In this treaty, the new United States gained the areas east of the Mississippi River, north of Florida, and south of Canada. Identify this treaty that ended the American Revolution.

(TREATY OF) PARIS (OF 1783)

23. At this book's climax, a policeman refuses to teach schoolchildren safety and a big accident happens as a result. Soon after, he returns to school assemblies with his funny canine assistant. Name this Caldecott winner by Peggy Rathmann.

OFFICER BUCKLE AND GLORIA

24. Some of these crafts created by certain Pacific Northwestern tribes are over 40 feet tall. They depict symbolic animals and spirits sacred to that clan or tribe. What are these large vertical logs?

TOTEM (POLE)

25. State your answer as a mixed number. There are 5 and a half cups of sugar in a container. If 2 and one-third cups of sugar are taken out to make cookies, how many cups of sugar are left in the container?

3 AND ONE-SIXTH (CUPS)

26. Correct the following sentence, if needed: Me and Tammy went to the movies.

CHANGE "ME AND TAMMY" TO "TAMMY AND I" (Accept equivalents)

27. In this agricultural method, furrows run crosswise to the slope, allowing the soil to absorb rainfall rather than wash away. It can reduce soil erosion by up to 50 percent compared to up and down hill farming. Give this term for plowing that follows the curves of hills or slopes.

CONTOUR (PLOWING) (Accept CONTOUR FARMING)

28. The first quarter of 2022 for a business lasts from the first day of January to the last day of March. Remembering that 2022 is not a leap year, how many days is this?

90 (DAYS)

29. This 1503 painting sits behind bulletproof glass in Paris's Louvre Museum. Its subject, probably the wife of a Florentine merchant, wears a mysterious smile. Name this portrait by Leonardo da Vinci.

MONA LISA (Accept LA GIOCONDA)

30. This property of matter is highly prized in printing paper. Name this property of matter, which means that light cannot shine through an object.

OPACITY (Accept other forms of the word such as opaque)

End of First Half – Begin Three-Minute Inquiry Period

Round 1 - Second Half

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1. A fort at this location was founded by Kentucky's most famous frontiersman. It is located along the Kentucky River in central Kentucky. Identify this location in Madison County, Kentucky, now home to a state park and national historic landmark.

(FORT) BOONESBOROUGH

2. For part of this novel, Valencia acts as a first person narrator. A limited third-person narrator gives the reader Virgil's, Chet's, and Kaori's perspectives about an incident of bullying. Name this Newbery-winning book by Erin Entrada Kelly.

HELLO, UNIVERSE

3. Philips Hue bulbs are examples of this type of light source, and they are more efficient and long lasting compared to traditional bulbs. They apply voltage to a semiconductor diode to make light. What is this type of light, more efficient than traditional incandescent bulbs?

LIGHT EMITTING DIODE (Accept LED)

4. This aspect of sheet music may resemble a top hat, the number 7, or a chicken's beak. Each one is held for the same number of beats as its corresponding note. What musical symbol tells a musician to remain silent?

REST

5. Using order of operations, what is the value of the numerical expression 3 plus 4 times 5 squared?

103

6. Cities in this Kentucky region include Owensboro, Henderson, and Beaver Dam. It is located in the west-central and northwestern part of Kentucky. What is this region of western Kentucky, named for a certain product mined there?

WESTERN COAL FIELD(S) (REGION)

7. How many prepositional phrases does the following sentence contain? The flock of Canadian geese flew over the trees by the river.

THREE (Accept the three phrases stated)

8. These clouds have horizontal layering and a uniform base. They are generally low in the atmosphere. Identify these clouds, contrasted with those of the cirrus or cumulus variety.

STRATUS (CLOUD(S))

9. This composer titled his only opera *Fidelio* [fih-DAY-lee-oh]. He may have dedicated the piano piece *Für Elise* to a soprano opera star. Identify this composer of *Moonlight Sonata*.

(LUDWIG VAN) BEETHOVEN

10. What is the least common multiple for 3, 8, and 12?

24

11. How many words in the following sentence need to be capitalized? My aunt lives on New York City's Upper West Side.

SEVEN (Accept the seven words stated)

12. Seas in this ocean include the Caribbean, the Mediterranean, and the Baltic. It is the second largest of the Earth's five oceans. Identify this ocean, which separates the Americas from Africa and Europe.

ATLANTIC (OCEAN)

13. The so-called "super" ones of these are often super cooled, and the "semi" ones include several metalloids. They are the opposite of insulators. Identify these materials that allow for easy flow of electricity and heat.

CONDUCTORS

14. This landmark sculpted by Frédéric Bartholdi [bar-TOLE-dee] has an interior designed by Gustave Eiffel. After being exposed to the elements, its covering of copper sheets developed a green patina [puh-TEE-nuh]. Give the name of this sculpture of a woman who guards New York Harbor.

(THE) STATUE OF LIBERTY (ACCEPT: LIBERTY ENLIGHTENING THE WORLD)

15. What is the name for the set of numbers that includes all of the natural numbers, zero, and the opposites of all of the natural numbers?

INTEGERS

16. Examples of this might be trying to overthrow your country's government or helping your country's enemies during wartime. Mary Surratt was convicted for this crime related to her involvement in the assassination of Abraham Lincoln. Give this term for betrayal of one's country.

TREASON

17. In later verses of this nursery rhyme, the title characters ride a pig and are attacked by a goat. The first stanza has them both falling down a hill, with the boy breaking his crown. What is this nursery rhyme about a failed attempt at fetching a pail of water?

JACK AND JILL

18. They consist of an anther and filament. They are the site where pollen is generated. Name this male reproductive structure in plants.

STAMEN(S)

19. This musical concept is the equivalent of ballet's pas de deux [pah du DYOO]. Give the word for a work played or sung by two people.

DUET

20. What is the result when negative 5 is multiplied by its reciprocal?

1

21. First and last name answer. Books written by her include *Women Who Work* and *The Trump Card*. She is the daughter of Ivana and wife of Jared. Who is this oldest daughter of Donald Trump?
- IVANKA TRUMP
22. An example of this sound device is, "Big black bears buy better books." Which sound device consists of nearby words that begin with the same consonant sound?
- ALLITERATION
23. These members of the earliest group of land vertebrates belong to the order Urodela and are not frogs, toads, or caecilians. What are these amphibians, that look like lizards?
- SALAMANDER(S) (Accept NEWTS)
24. This musical's characters include Old Deuteronomy, the Rum Tum Tugger, and Mr. Mistoffelees. Macavity and Grizabella are also members of the Jellicles, each of whom hopes to ascend to the Heaviside Layer. Name this Andrew Lloyd Webber musical about a group of felines.
- CATS
25. What is the solution to the equation $4X$ plus 10 equals 70?
- (X EQUALS) 15
26. John Winthrop was an influential leader of this religious group in the early 17th century. They were an industrious group of people, and virtually everything in their households was handmade. Identify this group of immigrants, English Protestants from the late 16th and 17th centuries.
- PURITAN(S)
27. Poems in this narrative genre may start in medias res [MAY-dee-ahs RAYS]--that is, in the middle of the action. They are long poems that feature larger than life heroes who go on quests or accomplish great things. *Beowulf*, *The Odyssey* and *Paradise Lost* are examples of what kind of poem?
- EPIC (POEM)
28. They are filter feeders and belong to the phylum porifera. What are these simple aquatic animals that are invertebrates, and can be used to make a cleaning tool?
- (SEA) SPONGE(S)
29. This niece of a Hollywood director choreographed *Fall River Legend* for the American Ballet Theatre. She also choreographed Aaron Copland's cowboy ballet *Rodeo*. Who choreographed the Broadway musicals *Brigadoon*, *Carousel*, and *Oklahoma!*?
- (AGNES) DE MILLE
30. What is the volume, in cubic units, of a cube if the length of a side of the cube is 3 units?
- 27 (CUBIC UNITS)

End of Second Half – Begin Three-Minute Inquiry Period

Extra Questions

- SS-1 Articles of clothing such as coats, hats, and boots are commonly produced through this old-fashioned activity. The harvest was completed through both trapping and hunting. What was this practice, associated with hunting of beavers, minks, and similar mammals?

FUR TRADING

- AH-2 One work by this man features 180 figures struggling to free themselves from swirling masses of material, a theme inspired by Dante's *Inferno*. Who is this sculptor of *The Gates of Hell*, better known for *The Thinker*?

(AUGUSTE) RODIN

- SC-3 This type of motion is seen when the hour hand on a grandfather clock makes a circle from noon to midnight. What kind of motion is also shown when Earth spins on its axis?

ROTATION (Accept other forms of the word)

- M-4 What is the numerator of the improper fraction when the mixed number 4 and three-fifths is rewritten as an improper fraction?

23

- LA-5 One member of this homophone pair means, quote, "to enter or escape through an opening usually by a fault or mistake." The other word is a vegetable similar to an onion, as well as the national emblem of Wales. What is the shared pronunciation of these two words?

LEAK/LEEK

